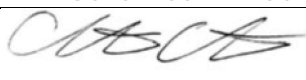


Burns – Thermal and Chemical

Guideline Title: Burns – Thermal and Chemical

Guideline Number: GUID-3203-T002

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 6/8/2021	Developed by: Air Care Team
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Department Numbers: 3203	

- I. **PURPOSE:** To maintain adequate oxygenation/ventilation, tissue perfusion, reduce potential for infection, provide adequate airway protection, and transport the burn trauma patient to an appropriate facility with burn treatment capabilities.

II. **PATHOPHYSIOLOGY:**

A burn wound is an injury caused by the interaction of an energy form (thermal, chemical, electrical, or radiation) and tissue. The depth of a thermal burn (partial or full thickness) is primarily defined by the temperature of the burning agent, the duration of exposure, and the conductance of the tissue involved. Long term outcomes are frequently dependent on the depth of the wounds and the total body surface area involved. In the acute setting, concomitant traumatic and inhalation injuries are a primary cause of mortality. There are three types of identifiable **inhalation injuries** including asphyxiation from carbon monoxide (CO) poisoning, supraglottic injury, which is primarily thermal in nature, and infraglottic injury which is primarily chemical.

Carbon monoxide binds with available hemoglobin to form carboxyhemoglobin and subsequently causes hypoxia. The signs and symptoms of carbon monoxide poisoning include tachycardia, tachypnea, headache, dizziness, nausea, pink to cherry red skin. Levels of 0% to 15% rarely cause symptoms and may be normal. Levels greater than 40% can cause mental obtundation and coma. Supraglottic injury can be suspected when facial burns, singed facial hair or arboraceous sputum are present. The provider must maintain a high index of suspicion based upon a history of exposure to high intensity heat in an enclosed space or inhalation of superheated air/stream. Infraglottic injury is progressive in nature and is caused by the inhalation of particulate by-products of combustion. It is manifested by an increased in both pulmonary vascular resistance and pulmonary capillary permeability, which causes pulmonary edema. The primary symptom is hypoxia that is resistant to oxygen therapy. Cyanide toxicity should be suspected in any victim of smoke inhalation who exhibits neurologic, respiratory, or cardiovascular compromise. Signs and symptoms of cyanide exposure include shortness of breath, headache, drowsiness, dizziness, tachycardia, tachypnea, perspiration, and anxiety. In higher levels, seizures, cardiac dysrhythmias, loss of consciousness, respiratory depression, and cardiovascular collapse can occur.

Chemical burns differ from thermal burns in that the burning process continues until the agent is inactivated by reaction with the tissues, is neutralized, or diluted with water. Dry chemicals such as lime should be brushed off prior to irrigation.

XCII. **DEFINITIONS:**

- A. CO – Carbon monoxide
- B. SBP – Systolic blood pressure
- C. TBSA – Total body surface area

XCIII. **DEPARTMENT GUIDELINE:**

- A. Assure scene safety.
- B. STOP the burning process and assure decontamination. This may require copious irrigation of the burn wound, as in the case of chemical burns, or simply removal of clothing or jewelry. Contact lenses should be removed.
 - i. For smaller size injuries <5%, cool burn briefly 3-5 minutes with water. Avoid ice or cold compresses, especially in a pediatric patient.
 - ii. Tar and asphalt burns are an exception to brief cooling. These products must be thoroughly cooled with copious amounts of cool water.
- C. All burn patients should have a full trauma assessment and be managed accordingly. Maintain spinal precautions if any question of trauma exists.
- D. Keep the patient warm. Apply blankets.
 - i. If burn has already been cooled, remove all wet dressings, and replace with a clean, dry covering.
- E. Assess and manage airway, breathing and circulation. If airway is involved, intubate early per *GUID-G008 Rapid Sequence Induction for Endotracheal Intubation* and *GUID-PR020 Endotracheal Intubation*. Indications for early intubation:
 - i. Signs of airway obstruction: hoarseness, stridor, accessory respiratory muscle use, sternal retraction
 - ii. Extent of the burn (TBSA burn > 40-50%)
 - iii. Extensive and deep facial burns
 - iv. Burns inside the mouth
 - v. Significant edema or risk for edema
 - vi. Difficulty swallowing
 - vii. Signs of respiratory compromise: inability to clear secretions, respiratory fatigue, poor oxygenation or ventilation
 - viii. Decreased level of consciousness where airway protective reflexes are impaired.
 - ix. “Flash Burns” to the face are less likely to result in airway edema that prolonged exposure to hot or irritant gasses in enclosed areas
- F. Administer high-flow oxygen via nonrebreather at 100%.
- G. Elevating the head of the patient’s bed will mitigate edema, especially in airway or facial burns.
- H. If bronchospasm present, administer **Albuterol 2.5mg** in 3ml of NS via nebulizer. Repeat as necessary per *GUID-PR012 – Inhalation Therapy*.
- I. Establish adequate vascular access. IVs/IOs may be placed in burned areas.
- J. Infuse crystalloids (warm if available) sufficient to prevent hypotension. The initial choice of fluid used in burn resuscitation is Ringer’s Lactate (LR).
 - i. Per ABLIS, prior to calculating the TBSA, the initial fluid rates for patients with visibly large burns are based on patient age:
 - 1. 5 years old and younger: 125ml LR per hour
 - 2. 6-13 years old: 250ml LR per hour

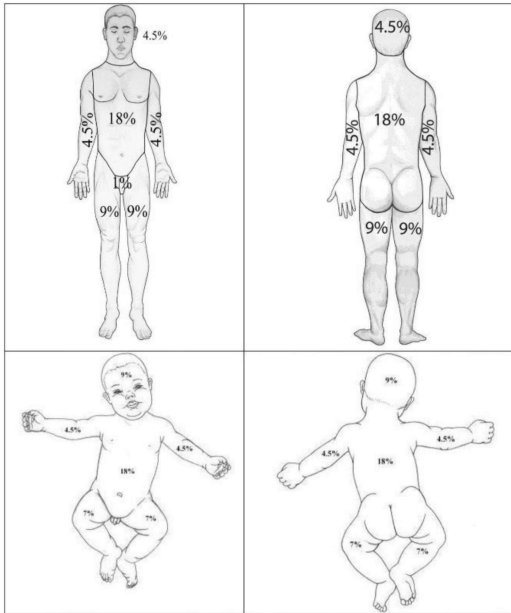
3. 14 years and older: 500ml LR per hour
- ii. Use the **Consensus fluid resuscitation formula** to estimate fluid requirements for partial/full thickness burns >20% TBSA for adults and older children 14 years and above.
 1. **2 ml LR x (weight in kgs) x (% TBSA) over first 24 hours.**
 2. **For children <14 years old: 3 ml LR x (weight in kgs) x (% TBSA)**
 3. **For Infants and young children <= 30kg: 3 ml LR x (weight in kgs) x (% TBSA) plus maintenance rate see below**
- iii. Give the first half over the first 8 hours from time of burn, and the second half over the remaining 16 hours.
- iv. An hourly urine output of 0.5-1 ml/kg of body weight for adults and for children 1-2 ml/kg can be indicative of appropriate volume resuscitation.
- v. For **pediatric patients** ≤ 30Kg, maintenance fluids with dextrose must be calculated and administered in addition to fluid resuscitation formula per *GUID-PED001 General Management of the Pediatric Patient*.
- K. Treat pain and/or anxiety per *GUID-G005 - Analgesia and Sedation Management and GUID-G006 Anxiety- Combative Patient Management*.
- L. Electrical burn patients should be monitored for life threatening dysrhythmias. Full spinal immobilization should be established. Check for entrance and exit wounds.
- M. Insert gastric tube and indwelling urinary catheter if doing so does not delay transport. If urinary catheter present, empty bag and begin measuring hourly urinary output.
- N. Wrap with dry clean sheet. Remove wet dressings if involved TBSA is >10%.
- O. In cases of significant burns, consider insertion of nasogastric tube per *GUID-PR010 - Gastric Tube Insertion*.
- P. In the absence of significant non-burn related trauma, transport to burn center if:
 - i. >10 % partial thickness burn
 - ii. Third-degree burns in any age group.
 - iii. Burns of hands, face, genitalia, feet, perineum, or major joints.
 - iv. Inhalation injury.
 - v. Chemical or electrical burns, including lightning injury.
 - vi. Circumferential burns of an extremity.
 - vii. Burns on patients with pre-existing medical illnesses that could complicate management, prolong recovery, or affect mortality.
 - viii. Burn injury that requires long-term rehabilitative support.
 - ix. Any patients with burns and concomitant trauma (such as fractures) in which the burn injury poses the greatest risk of morbidity or mortality. In such cases, if the trauma poses the greater immediate risk, the patient may be stabilized initially in a trauma center before being transferred to a Burn Center. Consult physician judgement.
- Q. Burn centers in the State of Florida include Orlando Health Orlando Regional Medical Center, Tampa General Hospital, Shands Health Care University of Florida – Gainesville, Jackson Memorial Hospital - Miami, Kendall Regional Medical Center - Miami, FL, and Blake Medical Center - Bradenton.
- R. If Cyanide poisoning is suspected treat per *GUID-T013 – Suspected Cyanide Toxicity*.

XCIV. **DOCUMENTATION:**

- A. EMS charts

XCV. **REFERENCES:**

Appendix



Extent of Burn

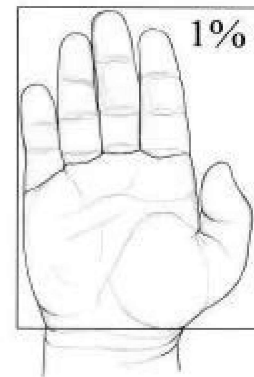
The most commonly used guide to estimate second and deeper degrees of burn is the “Rule of Nines.” In adults, distinct anatomic regions represent approximately 9% - or a multiple thereof – of the Total Body Surface Area (TBSA). In the infant or child, the “Rule” deviates because of the large surface area of the child’s head and the smaller surface area of the lower extremities. (Burn diagrams take factors into account.) Note that degree areas are not included in TBSA burn calculation.

Estimating Size for Scattered

The size of the patient’s hand – including the fingers – represents approximately one percent of total body surface area.

Therefore, using the patient’s

size as a guideline, the extent of irregularly scattered burns can be estimated.



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Burns

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Additional references for patient history based on ABLS Provider Manual

I. Circumstances: **Flame injuries**

- How did the burn occur?
- Did the fire occur inside or outside?
- Was the patient found inside a smoke-filled room?
- How did the patient escape? If the patient jumped out of a window, from what floor did he/she jump?
- Were others killed at the scene?
- Did the clothes catch on fire? How long did it take to extinguish the flames? How were the flames extinguished?
- Was gasoline or another fuel involved?
- Was there an explosion?
- Did the patient get thrown?
- Was the patient unconscious at the scene?
- Was there a motor vehicle crash? What was the mechanism of injury (T-bone, head-on, rollover other) How badly was the car damaged? Was there a car fire? Are there other injuries? Was the patient trapped in the burning vehicle? How long was he/she trapped?
- Is there any evidence of fuel or chemical spill that could result in a chemical burn as well as thermal injury?
- Are the purported circumstances of the injury consistent with the burn characteristics (i.e., is abuse a possibility)?

II. Circumstances: **Scalds**

- How did the burn occur?
- What was the temperature of the liquid?
- What was the liquid?
- How much liquid was involved?
- What was the thermostat setting of the water heater?
- Was the patient wearing clothes?
- How quickly were the patient's clothes removed?
- Was the burned area cooled? With what? How long?
- Who was with the patient when the burn took place?
- How quickly was care sought?
- Where did the burn occur (e.g., bathtub, sink)?
- Are the purported circumstances of the injury consistent with the burn characteristics (i.e., is abuse a possibility)? Pediatric scalds are sometimes due to child abuse. In addition to obtaining the patient history, it is helpful to ask EMS or other prehospital providers what they observed as the scene.

III. Circumstances: **Chemical injuries**

- What was the agent(s)?
- How did the exposure occur?
- What was the duration of contact?
- What decontamination occurred?
- Is there a Material Safety Data Sheet (MSDS) available?
- Is there any evidence of ocular involvement?
- Is there any evidence of illegal activity?

IV. Circumstances: **Electrical Injuries**

- What kind of electricity was involved – high voltage/low voltage, AC/DC?
- What was the duration of contact?
- Was the patient thrown or did he or she fall?
- Was there loss of consciousness?
- Was CPR administered at the scene?